

## 1.1 The Clinical History-Taking Bottleneck in Indian Hospitals

### PROBLEMS

patient's presenting complaints, history of present illness, past medical and surgical history, drug and allergy history, family and personal history, and a review of systems

### Ayurvedic

Ayurvedic history taking (Trividha, Ashtavidha, and Dashavidha Pariksha) requires detailed assessment of Prakriti (constitution), Vikriti (current imbalance), Agni (digestive capacity), Koshtha (bowel nature), Ahara-Vihara (diet and lifestyle), Nidana (causative factors), and Samprapti (pathogenesis)

## 1.2 The Documentation and Records Fragmentation Problem

point-of-entry mechanism to digitize, structure, and chronologically organize a patient's prior medical documents before they reach the consultation room.

### Ayushman Bharat Digital Mission (ABDM)

### ABHA (Ayushman Bharat Health Account) IDs

### FHIR

the 'first-mile' problem remains unsolved: there is no efficient, patient-facing software platform that captures structured history and digitizes documents into the ABDM ecosystem before the clinical encounter begins.

# SOLUTION/COMPARISON/WHAT THEY WANT

## 1.3 The Opportunity: AI-Powered Digital Clinical Intake Platform

1. AI-driven, multimodal clinical history acquisition with medical document digitization.
2. robust automatic speech recognition (ASR)
3. Indian languages and accents (Bhashini / AI4Bharat models)
4. large language models for conversational clinical history elicitation
5. high-accuracy OCR for handwritten and printed medical documents
6. ABDM's FHIR interoperability

## 2.1 The Problem in Precise Terms

software platform that enables patients to **independently** and **comprehensively** record their medical history through both **natural spoken** conversation and **guided touchscreen** interaction

**digitize** their existing physical medical documents, generating a structured, **physician-ready clinical history summary** that integrates with the **hospital information system (HIS)** and the ABDM ecosystem before the **patient enters the consultation room**.

## 2.2 Why Existing Solutions Fall Short

**Existing hospital registration systems** (currently deployed in some Indian hospitals) capture only demographic and appointment data— name, age, department, token number. They do not elicit any clinical history or process medical documents.

**Mobile health apps and tele-triage chat bots** require smartphone literacy, stable connectivity, and patient enrolment ahead of the visit - excluding the large elderly, rural, low-literacy, and first-visit patient populations who form the bulk of government hospital OPD load.

**Manual nurse-led triage/history desks** are themselves human-resource-limited, do not scale to 5,000+ daily patients, and reintroduce the same time and transcription bottleneck the system is trying to eliminate.

**Generic document scanners** digitize images but do not extract, structure, or

chronologically organize clinical content, nor link it to a structured history or ABHA record.

## 2.3 Specific Challenges a Solution Must Overcome

**Multilingual, multi-accent voice capture** in noisy hospital environments across Hindi, English, and major regional languages, for patients of varying literacy and digital comfort.

**Accessibility for low-literacy and elderly users** through intuitive icon-driven UI, audio prompts, and conversational guidance - the software platform must be usable by a first-time, non-tech-savvy patient with zero training.

**Accurate clinical history structuring** converting free-form patient narration into a standardized, physician-readable history (chief complaint, HPI, past history, drug/allergy, family, personal, review of systems) and, for AYUSH settings, Dashavidha Pariksha parameters.

**Reliable medical document digitization** OCR of handwritten and printed prescriptions, lab reports, and discharge summaries in multiple languages, with intelligent extraction of diagnoses, medications, and investigation values.

**Privacy, consent, and data security** compliance with the Digital Personal Data Protection Act 2023 and ABDM consent framework - handling sensitive health data within a secure software environment.

Expected solution:

3.1 Solution Overview - "MediKiosk" AI Clinical History Software Platform

The proposed solution - tentatively designated MediKiosk - a software platform for an AI-powered clinical history software platform

- 1. patient to record a comprehensive medical history through natural **voice conversation** and **guided touchscreen interaction**
- 2. **scan and digitize their existing physical medical documents**
- 3. generate a structured, **physician-ready clinical history summary** that is pushed to the **hospital information system (HIS)** and **linked to the patient's ABHA record**  
all completed before the consultation, with minimal staff assistance required.

3.2 Software Architecture

| Component         | Specification & Rationale                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Accessibility     | Audio-guided mode for visually impaired; large-text/high-contrast mode; multilingual UI; sign-language video avatar (stretch goal) |
| Patient dashboard | Personal login id and password for patients to put data and store medical records                                                  |

## Module A - Conversational Multimodal History Engine

A **conversational AI engine** that conducts a structured clinical history interview through both **voice and touch**. The patient speaks naturally in their preferred language; the engine asks **intelligent follow-up questions** (e.g., on stating 'chest pain', it probes onset, character, radiation, aggravating/relieving factors the **SOCRATES framework**) and simultaneously offers touch-based multiple-choice options for patients who prefer tapping. Built on Indian-language ASR, a dialogue manager constrained by a **clinical history ontology**, and **text-to-speech for audio prompts**.

**Adaptive questioning:** dynamically branches based on chief complaint and prior answers, mirroring a physician's clinical reasoning to elicit a complete HPI and review of systems

**Dual-mode input:** every question answerable by speaking OR tapping, ensuring usability across literacy and comfort levels

**AYUSH history mode:** for Ayurvedic OPDs, an extended interview capturing Dashavidha Pariksha (Prakriti, Vikriti, Sara, Samhanana, Pramana, Satmya, Sattva, Ahara Shakti, Vyayama Shakti, Vaya) and Ahara-Vihara assessment

**Red-flag detection:** AI flags emergency symptoms (e.g., acute chest pain with dyspnoea, stroke symptoms) and triggers immediate priority alert to triage staff rather than routine queueing

## Module B - Medical Document Digitization & Intelligence

An integrated scanning and document-AI pipeline that allows the patient to upload prior prescriptions, lab reports, and discharge summaries. The system performs high-accuracy OCR (printed and handwritten, multilingual), then extract and structure clinical entities.

**Intelligent extraction:** diagnoses, prescribed medications with dosages, investigation results with values and reference ranges, and procedure/surgery history

**Chronological organization:** automatically dates and orders documents into a coherent medical timeline for the physician

**Abnormal-value highlighting:** flags out-of-range lab values and potential drug interactions for physician attention

## Module C-Structured History Summary Generator

An AI summarization engine that synthesizes the conversational history and the digitized documents into a single, concise, physician-ready clinical summary in standard format presented on the consultation screen the moment the patient enters the room. The physician reads a complete, structured history in seconds rather than spending minutes eliciting it, and can edit/confirm before saving.

**Standard clinical format:** Chief complaint → HPI → Past medical/surgical → Drug & allergy Family → Personal → ROS → Prior investigations summary

**Editable & verifiable:** physician retains full control the summary is a draft to accept, amend, or reject, never an autonomous diagnosis

**Bilingual output:** patient-facing audio confirmation in local language; physician-facing summary in English/Hindi

## Module D - Consent, Privacy & ABDM Integration

A robust consent and security layer compliant with the Digital Personal Data Protection Act 2023 and the ABDM consent framework. The patient authenticates via ABHA ID, grants explicit consent for data capture and sharing, and the structured history is pushed to the hospital HIS/EMR and linked to the ABHA Personal Health Record via FHIR APIs.

**Secure processing:** voice and document AI are processed securely within the software platform

**Session termination:** temporary session data is cleared immediately after submission **Consent-first design:** granular, revocable consent with audio explanation for low-literacy patients

### 3.4 End-to-End Patient Journey

**Step 1 - Identify:** Patient logs into the software platform, enters/scans ABHA ID or Aadhaar details or registers as new; selects language; grants consent (audio-guided)

**Step 2 Converse:** AI conducts adaptive voice + touch history interview, capturing chief complaint, HPI, and full history; red flags trigger priority triage

**Step 3 Scan:** Patient uploads prior prescriptions, lab reports, and discharge summaries; AI digitizes, structures, and timelines them

**Step 4 Summarize & Route:** AI generates structured history summary, links to ABHA, pushes to HIS, updates the patient's digital record; summary appears on physician's screen at consultation

**Step 5 Consult:** Physician reviews complete history in seconds, edits/confirms, and devotes the full consultation to examination, reasoning, and counselling

